

# The Market Already Knew: A Pre-Registered Falsification of TimesFM-3 on the SPY Implied-Volatility Surface

Charlie Yan\*

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## Abstract

On the day of its release we subjected TimesFM-3, Google’s 330-million-parameter multivariate time-series foundation model, to a pre-registered falsification test on a task it has never seen: forecasting the SPY implied-volatility surface ( $25\Delta$  put, ATM,  $25\Delta$  call at 7/30/60/90-day constant maturities) at one- and five-day horizons — against random walk, AR(1), log-HAR, a 3-factor PCA-VAR, and the market’s own forward-variance-implied forecast, under pinball loss on its nine native quantiles, with Diebold–Mariano tests, a 90% Model Confidence Set, and a symmetric Mincer–Zarnowitz recalibration. Four results, in the order the kill ladder met them. *First*, the foundation models win the econometric contest outright: zero-shot, the four TimesFM configurations are the *only* members of the 90% MCS at  $h=1$  (pinball  $\times 10^3$  23.52–23.61 against 23.81 for log-HAR and 24.84 for the PCA-VAR; DM  $t = -13.7$ ), the gain is uniform across eleven years, and it survives in the Apache-licensed TimesFM-2.5 — to our knowledge the first evaluation of a time-series foundation model on option-implied volatility. Variate attention, the release’s headline capability, is worth nothing at one day and is what separates the models at five (46.00 against 46.41 univariate, which drops from the MCS). *Second*, most of the short-horizon edge is calibration, not information: recalibrated, the  $h=1$  MCS readmits the random walk and AR(1), the margin over persistence shrinking from 2.4% to 0.5% — replicating on *implied* volatility what Brini (2026) found on realized. *Third* — the kill — the market is the better forecaster wherever it quotes one: on ATM nodes the forward-curve-implied forecast beats every model at both horizons (0.02381 against 0.02437,  $p < 10^{-4}$ ; 0.04627 against 0.04741,  $p = 0.02$ ). The ladder stops here; no economic test is run; whatever variate attention learns about future ATM volatility, the term structure already prices. *Fourth*, one anomaly survives adversarial audit and then fails on licensing: on the  $25\Delta$  wings, where no market forecast exists, TimesFM-3’s five-day risk-reversal forecast retains information orthogonal to HAR, the PCA-VAR and trailing  $z$ -score fades jointly (partial IC +0.25; +0.15 after a first-day quote-bounce guard,  $2.8\times$  a conservative round-trip spread) — while the deployable TimesFM-2.5’s collapses from  $2.0\times$  to  $0.6\times$  under the same guard. The skill replicates across generations and licenses; the one tradeable-looking residual exists only in weights whose license forbids production use.

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\*v1.0 (SSRN submission draft), September 2026 — begun the day of the TimesFM-3.0 release. All results are SCREEN-basis forecast-loss comparisons on end-of-day quoted surfaces; no fill simulation and no performance claims. The full protocol — models, quantile recipe, walk-forward design, statistical tests, and four ordered kill criteria — was registered before any model ran, and every deviation is a dated amendment (Appendix A). Data construction and validation are in Section 2. TimesFM 3.0 weights are distributed under a non-commercial license; this study is research-only and disconnected from any trading infrastructure. Declaration of interest: the author independently trades equity index options, including SPY volatility strategies; the research was self-funded, with no funding from or affiliation with any model vendor. Code, pre-registration and the archived walk-forward loss series: <https://github.com/charlieyanhx/market-already-knew>.

**Keywords:** implied volatility surface; time-series foundation models; zero-shot forecasting; variate attention; forecast evaluation; model confidence set; Mincer–Zarnowitz recalibration; forward variance; pre-registration; options.

**JEL classification:** C45, C52, C53, G13, G17.

## 1 Introduction

Time-series foundation models promise zero-shot forecasts of any series, and their latest generation adds native multivariate structure: TimesFM-3 attends across series (“variate attention”) as well as across time, and accepts deterministic future covariates (TimesFM-3, 2026). An implied-volatility surface is a natural stress test of exactly that capability. Its twelve constant-maturity nodes are driven by a famously low-dimensional factor structure — level, slope and curvature explain most daily variation (Cont and da Fonseca, 2002; Gonçalves and Guidolin, 2006) — so a multivariate model must find structure *beyond* three linear factors to justify itself. And unlike realized volatility, the surface comes with a benchmark no statistical model enjoys: the market’s own term structure implies a forecast of its future ATM level under a martingale forward-variance assumption. A model that beats every econometric baseline but not the forward curve has discovered only what the market already prices.

This paper runs that test as a falsification exercise, registered in advance. We committed to the node set, the loss (pinball across TimesFM’s nine native quantiles), the model family, the walk-forward design, a uniform quantile recipe for point-forecast baselines, the statistical machinery (Diebold–Mariano with HAC errors; 90% Model Confidence Set; Mincer–Zarnowitz recalibration), and four *ordered* kill criteria with a stop rule, before any model ran. The prior art shaped the design. Goel et al. (2025) found TimesFM-2.0 could not consistently beat HAR on realized volatility zero-shot; Rahimikia et al. (2025) report uniformly negative zero-shot results on returns; and Brini (2026), in the most systematic comparison to date (nine foundation models, fifty assets), found that most of the short-horizon advantage was better-*scaled* forecasts rather than better prediction of dynamics — it vanished under Mincer–Zarnowitz recalibration except at the monthly horizon. Our design mirrors his key conventions (loss ratios averaged per series, never pooled; recalibration applied symmetrically) so that the implied-volatility answer is directly comparable to the realized-volatility one.

The falsification succeeds — the models die on the third criterion — but the autopsy is more informative than the verdict, and Sections 4–7 report all of it: a genuine and uniform win over the econometric family, including the 3-factor PCA–VAR, replicated across two model generations; a calibration decomposition that reproduces Brini (2026) on a new asset class; a defeat by the market’s own forward curve everywhere the market quotes a forecast; and one anomaly on the wings that survives an adversarial audit designed to kill it and then fails a licensing test.

## 2 Data and surface construction

**Chains.** We use freely available end-of-day SPY option chains, 2008-01-02 to 2025-12-12: 24,681,665 contract-day rows over 4,514 trading days. The vendor’s greeks fail internal-consistency checks (same-strike put and call implied volatilities disagree;  $\Delta_C - \Delta_P$  deviates materially from its parity value), so we discard them and recompute everything from quoted prices alone. Per date and expiry, a least-squares regression of  $C - P$  on strike over near-the-money pairs recovers the discount factor  $D$  and forward  $F$  (63,263 valid slices); one implied volatility per strike then comes from the OTM

leg by Black-76 bisection (5.89 million strike-level IVs), and deltas follow with parity satisfied by construction. Carry ( $r - q$ ) is taken spot-free from the ratio of parity forwards across two expiries, immune to the end-of-day timestamp mismatch between option and underlying closes.

**Nodes.** Within each expiry we interpolate IV linearly in recomputed delta to the  $25\Delta$  put, ATM (mean of the  $\pm 50\Delta$  interpolations), and  $25\Delta$  call; across expiries we interpolate linearly in total variance to 7/30/60/90-day constant maturities. No extrapolation anywhere: a node unbracketed in delta or tenor is missing. The result is 4,513 daily rows with all twelve nodes complete on 3,789 days (84%); the 7-day node requires the weeklies era and is sparse before 2011. Targets are  $\log(\text{IV})$  per node.

**Validation.** Against an independently sourced CBOE 16:15 snapshot grid (2020-09 to 2025-12 overlap, 1,320 days), the panel’s ATM-30d level correlation is 0.9985 (daily-change correlation 0.978), the  $25\Delta$  risk reversal 0.9935 (0.973), the  $10\Delta$  risk reversal 0.9975 (0.979), with level biases of 0.2–0.5 vol points — all above the 0.95 gate we set in advance. Notably, the same gate run on the vendor’s own greeks fails for the  $25\Delta$  risk reversal (0.906): the parity-forward recomputation is what makes the wings usable. One data artifact required a dated amendment: the source carries a systematic put–call quote shift that biases fitted discount factors upward at long maturities, and the sanity bound on  $D$  had to be widened (with the gate re-passed unchanged) to keep 100–140 DTE expiries available as the 90-day node’s upper bracket in the high-rate era.

### 3 Protocol

**Models.** Candidates: TimesFM-3 zero-shot, multivariate (all twelve nodes jointly) and multivariate with past-and-future covariates (actual FOMC, CPI and NFP release dates compiled from Federal Reserve and BLS primary sources; monthly and quarterly expiry flags; day-of-week; holiday adjacency), plus a univariate ablation with variate attention disabled. Controls: TimesFM-2.5 (Apache-2.0, the only deployable version); random walk; AR(1) on  $\log\text{-IV}$  changes; log-HAR (direct per horizon); a 3-factor PCA–VAR(1) — the primary registered benchmark — all refit on rolling 756-day windows each step; and the forward-curve-implied ATM forecast, which reads expected future ATM variance off the same day’s total-variance curve under a flat forward-variance assumption (defined only for ATM nodes; wings have no market-implied forecast).

**Loss and quantiles.** Primary loss is the pinball loss averaged over the nine deciles, on  $\log\text{-IV}$  at  $t+h$ ,  $h \in \{1, 5\}$  trading days, both horizons from a single model call. TimesFM’s native quantiles are used directly. Every point-forecast baseline is wrapped identically: its quantile  $q$  forecast is the point forecast plus the trailing empirical  $q$ -quantile of its own resolved  $h$ -step errors (252-day window, past-only) — no distributional assumption, and, as Section 4 verifies, empirically calibrated to within a point of nominal coverage.

**Walk-forward.** Forecast origins are every trading day; a forecast at origin  $t$  uses data through  $t$  only; the wrapper at  $t$  uses errors of origins resolved by  $t$ . The evaluation set per horizon is the intersection where every model has a full quantile vector, the current level is finite, and the target exists: 2,609 origins at  $h=1$  (2015-03-25 to 2025-12-11) and 2,565 at  $h=5$ . TimesFM receives the full available gap-free history as context at each origin (up to  $\sim 3,800$  days).

**Tests.** Per-origin surface losses (mean across the twelve nodes, which share the log scale) feed pairwise Diebold–Mariano tests with Newey–West errors and a 90% Model Confidence Set (Diebold and Mariano, 1995; Hansen et al., 2011); per-node loss ratios to the PCA–VAR are averaged across nodes, never pooled (Brini, 2026). The Mincer–Zarnowitz stage re-estimates, per model, node and

horizon, a rolling past-only (504-day) regression of realized on forecast, applies the affine map to all nine quantiles, and re-runs every test on the recalibrated losses (Mincer and Zarnowitz, 1969).

**Kill criteria** (ordered; stop at first failure): (1) TimesFM-3 multivariate fails to enter the 90% MCS against the PCA-VAR; (2) the edge disappears under Mincer-Zarnowitz recalibration; (3) the edge does not also beat the forward-curve-implied forecast on ATM nodes; (4) the edge does not survive a strictly post-release forward period. Criterion 4 is structurally unresolvable on release day; the archived loss series makes the re-score mechanical, and we commit to it.

## 4 Results: the econometric contest

Table 1 reports the headline panel. Four facts.

**The foundation models win the statistical contest outright, and they sweep it.** At  $h=1$  raw, the four TimesFM configurations take the top four places (23.52 to  $23.61 \times 10^{-3}$ ) ahead of log-HAR (23.81), AR(1) (24.13), the random walk (24.15) and the PCA-VAR (24.84), and they are the *only* members of the 90% MCS: every econometric model is excluded. The best rejects equality with the PCA-VAR at DM  $t = -13.7$ . At  $h=5$  the two multivariate TimesFM-3 configurations hold the MCS alone (46.00 and 46.05 against 47.48 for the PCA-VAR,  $t = -3.8$ ). Native quantiles are well calibrated out of the box (empirical coverage 0.092/0.509/0.900 at the 10/50/90 levels). Kill criterion 1 passes decisively.

**The result is not specific to the new model or its license.** TimesFM-2.5 — an earlier generation, 200M parameters, Apache-2.0 — also beats every econometric baseline at  $h=1$  (23.61 against log-HAR’s 23.81) and sits in the MCS with its successor. Whatever the architecture family is exploiting on this surface, it is not new to version 3 and not locked behind the non-commercial license. This matters for interpretation: the finding is about pretrained sequence models on quoted volatility, not about one checkpoint.

**Variate attention pays only at the longer horizon.** The univariate configuration — the same weights called once per node, so the model cannot look across the surface — is statistically indistinguishable from the multivariate one at  $h=1$  (23.54 against 23.56; both in the MCS, and the univariate is nominally ahead). At  $h=5$  the ordering separates: multivariate 46.00, univariate 46.41, and the univariate *drops out of the MCS* while both multivariate configurations remain. Cross-series structure is therefore not what makes these models good at one day — persistence and calibration are — but it is what distinguishes them at one week. This is worth stating because “multivariate foundation model” is the headline capability of the release, and on this surface it buys nothing at the horizon most people would test first.

**Covariates help marginally here, in contrast to realized volatility.** Adding the event calendar improves  $h=1$  raw loss from 23.56 to 23.52 and gives the only foundation configuration still in the MCS at  $h=5$  after recalibration — real but small. On realized volatility, a companion working paper by the author, run on the same underlying with the same release calendars, finds covariates to be the dominant effect, with covariate-bearing configurations beating univariate zero-shot by roughly  $8\sigma$ . The natural reading is that scheduled macro events matter for the *arrival of realized variance* far more than for the *quoted level* of implied variance, which already prices the calendar.

**The primary benchmark is itself beatable — horizon matters.** At  $h=1$  the 3-factor VAR is the *worst* model in the family, 4.5% worse than the random walk; at  $h=5$  it is the best econometric model. Factor mean-reversion earns its keep only at the weekly horizon. Studies that benchmark surface forecasts against a VAR at short horizons alone flatter their candidate.

Table 1: Track A (twelve raw nodes): mean pinball loss  $\times 10^3$  across nine quantiles, per-node loss ratio to the PCA-VAR (averaged across nodes, never pooled), Diebold–Mariano  $t$  against the PCA-VAR on per-origin surface losses (negative = better), and 90% MCS membership. “MZ” rows re-score all models after rolling past-only Mincer–Zarnowitz recalibration. 2,609 origins at  $h=1$ , 2,565 at  $h=5$ , 2015–2025.

| Stage | Model                | $h = 1$      |       |        |            | $h = 5$      |       |        |            |
|-------|----------------------|--------------|-------|--------|------------|--------------|-------|--------|------------|
|       |                      | Loss         | Ratio | DM $t$ | MCS        | Loss         | Ratio | DM $t$ | MCS        |
| raw   | TimesFM-3 MV+cov     | <b>23.52</b> | 0.930 | −12.2  | <b>yes</b> | 46.05        | 0.961 | −3.6   | <b>yes</b> |
| raw   | TimesFM-3 univariate | 23.54        | 0.931 | −13.1  | <b>yes</b> | 46.41        | 0.965 | −2.6   | no         |
| raw   | TimesFM-3 MV         | 23.56        | 0.932 | −13.7  | <b>yes</b> | <b>46.00</b> | 0.960 | −3.8   | <b>yes</b> |
| raw   | TimesFM-2.5          | 23.61        | 0.935 | −12.4  | <b>yes</b> | 46.66        | 0.970 | −2.0   | no         |
| raw   | log-HAR              | 23.81        | 0.943 | −12.4  | no         | 47.81        | 0.995 | +1.0   | no         |
| raw   | AR(1)                | 24.13        | 0.955 | −5.9   | no         | 49.75        | 1.028 | +3.4   | no         |
| raw   | Random walk          | 24.15        | 0.955 | −6.1   | no         | 49.64        | 1.026 | +3.3   | no         |
| raw   | PCA-VAR (3f)         | 24.84        | 1.000 | —      | no         | 47.48        | 1.000 | —      | no         |
| MZ    | TimesFM-3 MV+cov     | <b>23.78</b> | 0.924 | −11.9  | yes        | <b>47.64</b> | 0.940 | −3.9   | yes        |
| MZ    | TimesFM-3 MV         | 23.83        | 0.926 | −12.7  | yes        | 47.96        | 0.947 | −3.3   | no         |
| MZ    | TimesFM-2.5          | 23.85        | 0.927 | −11.5  | yes        | 48.32        | 0.950 | −2.7   | no         |
| MZ    | TimesFM-3 univariate | 23.85        | 0.927 | −11.8  | yes        | 48.56        | 0.955 | −2.2   | no         |
| MZ    | AR(1)                | 23.90        | 0.929 | −11.6  | yes        | 48.35        | 0.947 | −2.3   | yes        |
| MZ    | Random walk          | 23.92        | 0.929 | −11.9  | yes        | 48.40        | 0.948 | −2.3   | yes        |
| MZ    | log-HAR              | 24.03        | 0.934 | −12.9  | yes        | 50.45        | 0.993 | +1.0   | no         |
| MZ    | PCA-VAR (3f)         | 25.25        | 1.000 | —      | no         | 50.00        | 1.000 | —      | no         |

**The gain is uniform, not episodic.** Brini (2026) found foundation-model gains on realized volatility concentrated in a few outlier assets. The analogue here would be concentration in a few calendar episodes, and it does not occur: the yearly loss ratio of TimesFM-3 multivariate to the PCA-VAR at  $h=1$  raw has worst year 0.987 and full sample 0.948, with the ex-2020 (COVID) figure at 0.950, indistinguishable from the full sample. Uniformity also answers the standard objection to zero-shot claims on historical data — that pretraining may have absorbed correlated series. The 2024 and 2025 ratios (0.925 and 0.958) lie entirely after every disclosed pretraining cutoff in the model’s corpus (Wikipedia pageviews to late 2023, Google Trends to 2022) and match the full sample: whatever the model is exploiting, it exploits it equally on data that postdates everything it could have memorized.

## 5 Calibration, not information: the Mincer–Zarnowitz stage

The MZ rows of Table 1 change the interpretation without changing the ranking. After a rolling, past-only affine recalibration applied to every model symmetrically, the TimesFM configurations remain first and still reject against the PCA-VAR ( $t = -11.9$  for the best at  $h=1$ ) — so the registered criterion 2, stated against the primary benchmark, passes. But the 90% MCS now admits the recalibrated AR(1), random walk and log-HAR at  $h=1$ : the best model’s margin over simple persistence collapses from 2.4% to 0.5% and is no longer statistically separable. Against the models that matter — persistence, not the VAR — the short-horizon foundation-model edge on implied volatility is mostly a better-scaled forecast, precisely the pattern Brini (2026) documents on realized volatility, now replicated on a different volatility object with a different loss. What survives

recalibration is the beating of the PCA–VAR (which MZ cannot repair, since its deficit is dynamic misspecification, not scale) and a residual edge at  $h=5$ , where the covariate configuration is the only foundation model left in the MCS.

A second decomposition points the same way and sharpens it. Because both horizons come from a single model call, the implied day-two-to-five change (the  $h=5$  point forecast minus the  $h=1$ ) can be scored against the realized change from  $t+1$  to  $t+5$  — a construction that excludes the first day of the horizon entirely, and with it any component that is really cross-sectional denoising of day- $t$  marks rather than forecasting. Beyond day one the econometric contest closes (2,946 origins): the advantage over log-HAR loses significance (DM  $t = -1.2$ ) and the PCA–VAR is nominally ahead (change-IC +0.274 against +0.255). What survives past day one is the architecture ordering — multivariate beats the univariate ablation ( $t = -4.2$ ) and TimesFM-2.5 ( $t = -2.7$ ) — and the wing residual of Section 7, which was audited with precisely this guard. The registered full-horizon headline stands, but its anatomy is now explicit: the family-wide edge is front-loaded into the first day, where genuine one-day information and end-of-day mark noise cannot be separated on a quoted panel, while the cross-series information is what persists deeper into the horizon.

## 6 The kill: the market’s own forecast

Where the market quotes a forecast, it wins, and it wins against every model we ran. On the ATM subset (four nodes, 2,616 origins at  $h=1$ ), the forward-curve-implied forecast — expected future ATM variance read off the same day’s total-variance curve, with trailing empirical error quantiles like every other point model — posts 0.02381 against the best candidate’s 0.02437 at  $h=1$  (DM  $t = +4.08$ ,  $p < 10^{-4}$ ) and 0.04627 against 0.04741 at  $h=5$  ( $t = +2.32$ ,  $p = 0.02$ ), and beats the econometric models by wider margins still ( $t$  between +5.3 and +11.3 at  $h=1$ ). The result survives the MZ stage on both sides, and no configuration — multivariate, covariate-bearing, or the deployable 2.5 — comes closer than four standard errors at one day. Kill criterion 3 therefore fires, the ladder stops, and no economic test is run. This verdict has a twenty-year ancestor reached by the opposite route: Gonçalves and Guidolin (2006) found the surface’s VAR-factor dynamics strongly predictable and then ran the trading test we do not — statistical predictability failed to survive sensible transaction costs, and they declined to read their result as evidence against market efficiency; Bernales and Guidolin (2014) extended the same predictability-plus-economic-value program. Criterion 3 is the friction-free version of that test: where they asked whether the forecast’s profits outrun the toll, we ask whether the forecast outruns the price itself, and a model that cannot beat the quote has nothing left for the toll to consume. The information the foundation models extract about future ATM volatility — genuinely more than any econometric baseline here extracts — is a subset of what the term structure already prices. For a volatility book the practical reading is sharp: the value of the term structure’s forecast is captured by *trading* the curve, not by out-forecasting it at short horizons.

## 7 The wing anomaly, and why it is not deployable

The forward curve exists only at ATM. On the eight  $25\Delta$  wing nodes the candidates beat every baseline per node (raw DM  $t$  against the random walk between  $-2.7$  and  $-7.8$  at  $h=5$ ), and this is where a post-hoc follow-up — registered as such, with its bar fixed before the numbers were computed — found the one result that resists dying.



Define the model’s five-day  $25\Delta/30d$  risk-reversal forecast (put-wing minus call-wing predicted change, in vol points). Its gross IC against the realized change is +0.43 for TimesFM-3 multivariate; but gross is not the question, because a trailing  $z$ -score fade of the risk reversal — the classic skew mean-reversion signal — already achieves IC +0.35 to +0.37 on the same panel, and a signal that rediscovers skew fade proposes nothing. The registered test is therefore *orthogonal*: residualize the model’s forecast on HAR, the PCA-VAR, and 63- and 252-day trailing- $z$  fades jointly, and score the residual. It retains IC +0.248 ( $4.7\times$  a conservatively measured 0.17-vol-point round-trip spread at  $2\sigma$  signal strength). Two adversarial audits follow. At  $h=1$  the same construction collapses to IC +0.050 — below the bar; the one-day wing edge was comparators plus bounce. At  $h=5$ , excluding the first day entirely (scoring the residual against the change from  $t+1$  to  $t+6$ , so that any component that is really cross-sectional denoising of today’s marks earns nothing) cuts the IC to +0.147 — still  $2.8\times$  the spread. About 40% of the orthogonal edge was quote-bounce; the rest is structure that persistence, linear factors and skew fade jointly do not span.

**The deployability test fails.** TimesFM-3 weights are non-commercial, so the anomaly can only matter if the Apache-licensed TimesFM-2.5 carries it. It does not. TimesFM-2.5’s jointly-orthogonal wing edge is  $2.0\times$  spread before the bounce guard and  $0.63\times$  after it — the same audit that leaves TimesFM-3 at  $2.8\times$  (and the covariate configuration at  $2.5\times$ ) removes essentially all of 2.5’s. Table 2 collects the three candidates. The asymmetry is the paper’s sharpest practical point: the *surface-level* skill replicates across generations and licenses, while the one residual that looks tradeable exists only in weights whose license forbids production use. We report it as an anomaly for future work — it is post-selection, it is a screen on end-of-day marks rather than a backtest, and criterion 4’s forward re-score applies to it with full force.

Table 2: Five-day  $25\Delta/30d$  risk-reversal forecast: information coefficient against the realized change, gross and orthogonal to {log-HAR, PCA-VAR, 63d and 252d trailing- $z$  fades} jointly, before and after excluding the first day as a quote-bounce guard. Amplitude is IC  $\times 2\sigma$  of the realized change, expressed as a multiple of a 0.17-vol-point round-trip spread; the pre-registered bar is  $2\times$ .

| Model                         | IC gross        | IC orthogonal | IC orth., skip-1 | amp/spread (skip-1)  |
|-------------------------------|-----------------|---------------|------------------|----------------------|
| TimesFM-3 MV                  | +0.427          | +0.248        | +0.147           | <b>2.83</b> $\times$ |
| TimesFM-3 MV+cov              | +0.396          | +0.184        | +0.129           | <b>2.49</b> $\times$ |
| TimesFM-2.5 (deployable)      | +0.358          | +0.107        | +0.032           | 0.63 $\times$        |
| trailing- $z$ fade (63d/252d) | +0.347/ + 0.371 | —             | —                | —                    |

## 8 Costs, Track B, and specification accounting

**Inference cost, and a warning about sustained runs.** On an Apple M1 (Metal GPU, float32, 8 GB) one multivariate twelve-node forecast with full history costs 2.0s for TimesFM-3, 2.2s with covariates and 3.0s for TimesFM-2.5 — three orders of magnitude above any baseline here, since the entire 4,513-day PCA-VAR walk-forward refits in under 3s. Two engineering effects matter more than the headline number for anyone replicating this. First, naively varying context lengths forces per-shape GPU kernel recompilation, which degraded throughput  $8\times$  until context lengths were bucketed to a fixed grid. Second, and less obvious, *sustained* walk-forwards degrade catastrophically on consumer hardware: the covariate configuration that took 2.2 hours on Track A took **19.5 hours** for the same number of origins on Track B after many hours of continuous GPU work, under system-wide memory pressure. The signature identifies the cause — the p90 per-forecast time stayed

at 2.84s while the mean rose to 21.16s, so the loss stalls (paging), not compute. A zero-shot foundation model is cheap per call and expensive per campaign, and benchmark timings taken on a fresh machine will understate a multi-thousand-origin study by an order of magnitude. Results are device-independent: CPU and Metal forecasts agree to  $3 \times 10^{-7}$  on both point and quantile outputs.

**Track B (SSVI).** The registered arbitrage-free track — forecast daily SSVI parameters instead of nodes — failed its own pre-registered fit gate: a strict single- $(\rho, \eta, \gamma)$  power-law SSVI reconstructs the twelve nodes with median error 2.03 vol points against a 1.0 bar, so the kill ladder ran entirely on Track A, as the registration’s contingency specified, and Track B is reported as supplementary. Within it the ordering of Table 1 reproduces and sharpens: at  $h=1$  raw the covariate configuration leads at 0.2295 against 0.2414 for the random walk and 0.2895 for the PCA-VAR (DM  $t = -18.2$ ), and the econometric family is beaten by margins far wider than on the raw nodes — SSVI parameter series are simply harder for AR, HAR and a factor VAR to forecast. Those losses are computed on transformed parameters and are *not* on the same scale as Track A’s log-IV losses, so the two tracks’ levels are not comparable and we draw no arbitrage-free-versus-raw accuracy conclusion; Gate B0’s failure is what forecloses that comparison. What does carry across is the mechanism result: the univariate configuration sits in the MCS at  $h=1$  and drops out at  $h=5$  on *both* tracks, so the horizon-conditionality of variate attention is not an artifact of the node construction. The TimesFM-2.5 control on Track B was not run (amendment A7): it bears on no kill criterion, and after the 19.5-hour stage above the cost was judged disproportionate.

**Specification count.** Nine registered models, two tracks, two horizons, two stages, plus one registered post-hoc follow-up with five comparators and two audits; every configuration run — including the failed SSVI gate, a withdrawn univariate implementation that proved to be a silent no-op, the omitted Track B control, and every restart of the inference queue — is logged in the registration’s amendment ledger, which is the count any later multiple-testing correction must use.

## 9 Related work

Brini (2026) is the closest antecedent and the design template: nine zero-shot foundation models against the HAR family on realized volatility, with MCS and MZ stages; his headline — short-horizon gains are calibration; architecture dispersion exceeds the foundation-versus-econometric gap — is what our criterion 2 imports. Goel et al. (2025) and Rahimikia et al. (2025) report the earlier negative zero-shot results for TimesFM-2.0 on realized volatility and for foundation models on returns. On implied-volatility surfaces, the deep-learning literature forecasts nodes or factor dynamics with ConvLSTMs (Medvedev and Wang, 2021), arbitrage-penalized networks (Zhang et al., 2021), SVI-parameter dynamics (Bloch and Böök, 2021), diffusion models (Jin and Agarwal, 2026), and functional time series (Shang and Kearney, 2021), over the classical factor-VAR tradition (Gonçalves and Guidolin, 2006; Cont and da Fonseca, 2002); none, to our knowledge, evaluates a pretrained foundation model, and — more surprisingly — none benchmarks against the market’s own forward-variance forecast, the tradition of Bollerslev et al. (2010). Our kill criterion 3 is that benchmark, and it is the one no candidate could pass. Two scope limits deserve statement rather than burial. We evaluate one model family; Brini (2026) finds that dispersion *across* foundation-model architectures exceeds the foundation-versus-econometric gap on realized volatility, and nothing here speaks to whether another architecture would fare differently against the forward curve. And our horizons stop at one week; the one place Brini (2026) found a genuine post-recalibration informational gain was the monthly horizon, which our registered design does not reach.



## 10 Conclusion

Read as a horse race, this is a strong entry beaten. On the day of its release, TimesFM-3 out-forecast every econometric baseline we could field on a twelve-node implied-volatility surface — uniformly across eleven years, with honestly calibrated quantiles, replicating across two model generations and both licenses — and still failed the only two questions a trading desk would ask. Is the edge information rather than calibration? Mostly not, at one day. Does it know anything the market has not already priced? Not wherever the market publishes a forecast of its own.

Read properly, the race is an efficiency measurement, and the interesting subject is not the model but the price. Release-day timing makes the measurement clean: no market participant could yet have been trading on this model, and the forward curve beat it anyway, at four standard errors. A zero-shot checkpoint is, moreover, a public good — whatever a downloadable model knows about future volatility is destined for the price as adoption spreads, so its edge is self-erasing in a way a private signal’s is not. Our one surviving anomaly obeys that logic with almost comic precision: the five-day wing residual that clears every adversarial audit at  $2.8\times$  spread lives only in the checkpoint whose license forbids production use — the market is under no obligation to absorb what nobody may deploy — and vanishes in the deployable one. [Gonçalves and Guidolin \(2006\)](#) concluded from 1990s data that surface predictability was not evidence against market efficiency; twenty years and some seven orders of magnitude of parameter count later, that conclusion survives with the friction removed. The strongest sequence model yet pointed at this surface met a better forecaster on its first morning, and it was the quote.

Whether the wing residual is a fact about model capacity or a warning about how thin such residuals really are, the strictly-post-release period that neither we nor the models have seen will decide; that re-score is committed, mechanical on the archived loss series, and the next pre-registration.

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## A Registration summary

Registered 2026-08-31 (release day), before any model ran: node set and construction; log-space targets; both horizons; the nine-model family; pinball across nine deciles as primary loss; the uniform trailing-empirical quantile wrapper (252-day window); rolling 756-day refits; DM/NW, 90% MCS, rolling 504-day past-only MZ; per-node ratio averaging; the four ordered kill criteria with stop rule; and the expectation, stated as a prior, that the most likely outcome was a kill at criterion 1 or 2. Dated amendments (all before any candidate-versus-baseline number existed for the affected component): A1 widened the parity discount-factor bound for the free source’s long-maturity quote-shift artifact; A2 recorded the SSVI fit-gate failure and demoted Track B to supplementary; A3 dropped the sparse pre-2017 7-day theta from Track B’s series; A4 implemented the univariate configuration as a variate-attention ablation after the strict per-series path proved computationally infeasible ( $\sim 21$  s/forecast); A5 and W3b fixed the wing follow-up’s comparator set and bar (orthogonal increment over the best of HAR/PCA-VAR/trailing- $z$ , amplitude/spread  $\geq 2$  at  $h=5$ ) before the partial ICs were computed. The walk-forward loss series for every model, track, horizon and stage is archived for the criterion-4 forward re-score.